

Model	CODE				SUMM		SIMP		BIO		R-BIN		R-NUM		
	Avg U $\uparrow$	Avg H $\downarrow$	Avg R $\uparrow$	Utility	H/R	Utility	H/R	Utility	H/R	Utility	H/R	Utility	H/R		
Alpaca 7b	0.46	0.52	0.95	0.96	0.0/0.96	0.3	0.7/1.0	0.69	0.31/1.0	0.28	0.61/0.72	0.45	0.55/1.0	0.06	0.94/1.0
Falcon 40b instruct	0.61	0.37	0.95	0.93	0.06/1.0	0.77	0.14/0.9	0.85	0.13/0.98	0.5	0.5/1.0	0.25	0.71/0.87	0.33	0.66/0.98
GPT-3.5	0.70	0.3	1.0	0.94	0.06/1.0	0.98	0.02/1.0	0.94	0.06/1.0	0.83	0.17/1.0	0.17	0.83/1.0	0.34	0.66/1.0
GPT-4	0.70	0.29	0.99	0.96	0.04/1.0	0.97	0.03/1.0	0.95	0.05/1.0	0.82	0.13/0.95	0.14	0.86/1.0	0.37	0.63/1.0
Llama-2 7b chat	0.64	0.35	0.99	0.92	0.06/0.98	0.96	0.04/1.0	0.91	0.09/1.0	0.47	0.51/0.95	0.43	0.57/1.0	0.17	0.83/0.99
Llama-2 13b chat	0.66	0.34	1.0	0.93	0.07/0.99	0.96	0.03/1.0	0.91	0.09/1.0	0.49	0.51/1.0	0.42	0.58/1.0	0.22	0.78/1.0
Llama-2 70b chat	0.6	0.36	0.94	0.93	0.06/1.0	0.97	0.03/1.0	0.93	0.07/1.0	0.43	0.34/0.65	0.16	0.84/1.0	0.19	0.81/0.99
Llama-3 8b chat	0.58	0.4	0.97	0.92	0.05/0.97	0.95	0.04/0.99	0.89	0.1/0.99	0.48	0.45/0.87	0.11	0.89/1.0	0.14	0.86/1.0
Llama-3 70b chat	0.65	0.34	0.99	0.94	0.06/1.0	0.98	0.02/1.0	0.92	0.08/1.0	0.64	0.35/0.98	0.12	0.87/0.93	0.31	0.69/1.0
Mistral 7b instruct	0.61	0.37	0.97	0.91	0.02/0.92	0.94	0.06/1.0	0.9	0.1/1.0	0.48	0.52/0.99	0.21	0.79/1.0	0.22	0.75/0.9
Mixtral 8x7b instruct	<u>0.68</u>	0.32	0.99	0.94	0.06/1.0	0.96	0.04/1.0	0.92	0.08/1.0	0.67	0.33/1.0	0.22	0.77/0.96	0.34	0.65/1.0
OLMo 7b instruct	0.55	0.44	0.99	0.93	0.06/1.0	0.91	0.09/1.0	0.86	0.14/1.0	0.37	0.62/0.98	0.13	0.87/1.0	0.13	0.87/0.98
Redpajama 3b chat	0.58	0.42	1.0	0.96	0.04/1.0	0.84	0.16/1.0	0.63	0.37/1.0	0.32	0.68/1.0	0.61	0.39/1.0	0.14	0.86/1.0
Redpajama 7b chat	0.44	0.56	1.0	0.95	0.05/1.0	0.53	0.46/0.99	0.53	0.47/1.0	0.31	0.69/1.0	0.19	0.81/1.0	0.1	0.9/1.0

Table 2: Model performance on **HALOGEN** task sets for **Response-Based** categories: code, text summarization, text simplification, biographies, rationalizations-binary and rationalizations-numerical. For each set, we report the average utility of model responses, as well as corresponding hallucination scores/response ratios. The top result is highlighted in green, and the second-best in orange.

Model				References		Historical Events		False Presuppositions	
	Avg Utility $\uparrow$	Avg H $\downarrow$	Avg R $\downarrow$	Utility	H/R	Utility	H/R	Utility	H/R
Alpaca 7b	0.47	0.88	0.53	0.97	0.72/0.03	0.13	1.0/0.87	0.3	0.91/0.7
Falcon 40b instruct	0.21	0.87	0.79	0.26	0.74/0.74	0.22	1.0/0.78	0.16	0.88/0.84
GPT-3.5	0.64	0.76	0.36	0.33	0.62/0.67	0.96	1.0/0.04	0.62	0.68/0.38
GPT-4	0.71	0.66	0.29	0.52	0.33/0.48	1.0	1.0/0.0	0.61	0.65/0.39
Llama-2 7b chat	0.56	0.87	0.44	0.18	0.76/0.82	1.0	1.0/0.0	0.5	0.87/0.5
Llama-2 13b chat	0.33	0.88	0.67	0.2	0.75/0.8	0.73	1.0/0.27	0.05	0.88/0.95
Llama-2 70b chat	0.46	0.88	0.54	0.19	0.77/0.81	1.0	1.0/0.0	0.2	0.88/0.8
Llama-3 8b chat	0.55	0.81	0.45	0.23	0.63/0.77	0.93	1.0/0.07	0.48	0.8/0.52
Llama-3 70b chat	0.57	0.76	0.43	0.27	0.56/0.73	1.0	1.0/0.0	0.45	0.74/0.55
Mistral 7b instruct	0.41	0.86	0.59	0.24	0.78/0.76	0.32	1.0/0.68	0.67	0.8/0.33
Mixtral 8x7b instruct	0.36	0.82	0.64	0.23	0.59/0.77	0.65	1.0/0.35	0.19	0.87/0.81
OLMo 7b instruct	0.32	0.87	0.68	0.05	0.75/0.95	0.34	1.0/0.66	0.57	0.85/0.43
Redpajama 3b chat	0.16	0.86	0.84	0.11	0.7/0.89	0.37	1.0/0.63	0.01	0.87/0.99
Redpajama 7b chat	0.26	0.84	0.74	0.14	0.61/0.86	0.49	1.0/0.51	0.16	0.92/0.84

Table 3: Model performance on **HALOGEN** task sets for **Refusal-Based** categories: scientific attribution, historical events, and false premises. For each set, we report the average utility of model responses, as well as the corresponding hallucination scores/response ratios for models on that set. The top result is highlighted in green, and the second-best in orange.

well as training data. We find that on both response-based and refusal-based tasks, GPT-3.5 and GPT-4 (closed-source models) are currently clear winners, suggesting room for improvement for open models. Amongst the open-source models, Llama-3-70B demonstrates the best performance.

Do larger models hallucinate less? We find that on response-based tasks, larger models generally hallucinate lesser than smaller models, as demonstrated by lower hallucination rates on four out of six tasks ($\text{LLAMA-2 70B} \leq 13\text{b} \leq 7\text{b} / \text{LLAMA-3 70B} \leq 8\text{b}$). On refusal-based tasks, we do not observe a similar trend. Further, we find that Mixtral 8x7b (a MoE model, with 7B active parameters) hallucinates less than MISTRAL 7B on average, in both response-based and refusal-based settings.

5 Why Do Models Hallucinate?

Armed with an extensive dataset of model hallucinations, we seek to gain a understanding of potential sources of model hallucination— by tracing

back model hallucinations to pretraining corpora. We isolate individual hallucinated atomic facts and assign error classes of the following types:

- **Type A:** The correct fact was present in the pretraining data.
- **Type B:** An incorrect fact was in the pretraining data, or the fact is taken out of context i.e. the fact appeared within a specific setting in a document in the training data, but when taken in isolation, it loses its original meaning.
- **Type C:** Neither a correct nor an incorrect fact was present in the pretraining data, and the model over-generalized when making predictions.

It is possible for a model response to have both Type A + Type B errors, when the pretraining data contains both incorrect and correct facts—for instance, a pretraining corpus could include factually accurate news articles indicating that Barack

Obama was born in Hawaii, along with conspiracy theory websites falsely asserting he was born in Kenya. For content-grounded tasks, there is a fourth source: models generating inferences not supported by the provided context; see §5.2.

5.1 Open-Ended Tasks

Code We shed light on large language model hallucinations when generating software packages. We extract hallucinated packages for 8 models: OLMo, Llama-2-7B/13B/70B, Llama-3 8B/70B and GPT-3.5/4. Of these models, only OLMo is accompanied by public disclosure of its training data. For the Llama family, we consider C4 as a potential source (Raffel et al., 2020; Touvron et al., 2023a), and for GPT-3.5/4 we consider OpenWebText (Gokaslan and Cohen, 2019).

We find that across models, **hallucinated software packages can be found in pretraining corpora to a large extent** (Table 4)—in one case up to $\sim 72\%$ of hallucinated packages appear to be drawn from pretraining corpora (**Type B error**). To understand better the contexts these packages appear in, we qualitatively examine matched documents for five packages hallucinated by each of the models. We find several potential sources of error for hallucinated packages that appear in the training data, including: (a) the hallucinated package is a local import within a repository or codebase, (b) the hallucinated package has a different name in the package index, (c) the hallucinated package is deprecated, (d) the hallucinated package is actually a class or a function within another package, and (e) the hallucinated package appears in the context of a non-Python program.

Historical Events We analyze model hallucinations in instances where models hallucinated meetings between historical figures. For models which have at least 100 hallucinations in this category (OLMo, Llama-2 13b, Llama-3 8b), we sample 100 instances and categorize hallucinations by computing co-occurrence statistics in pretraining corpora based on the following schema: (1) Type A errors: birth and death date of both the entities are in training corpora, in the same document as the entity, (2) Type B: both entity names occur in a single document in the pretraining dataset, (3) Type C: the birth date and death date of either of the entities does not occur in the same document with the entity name in the pretraining corpora. We find that for all three models, the entity names rarely co-occur in

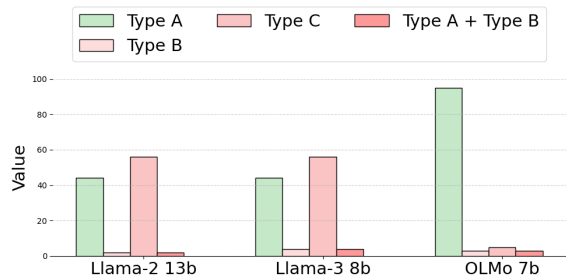


Figure 3: **The counts of types of model hallucinations when describing hypothetical historical events.** Models seldom make Type B errors, indicating there is unlikely to be basis in pretraining data.

the same document, indicating that the model may not have documents in pretraining data that lend supporting evidence to the hallucination (Figure 3).

Senator Search We analyze hallucinations in cases where models predict incorrect educational affiliations for senators. We analyze 500 instances for Llama-2 7B/13B/70B, Llama-3 8B/70B and OLMo. We also extract the correct educational affiliations of senators from Wikidata. We categorize hallucinations as: (1) Type A errors: A Wikipedia article containing the correct educational affiliation is present, (2) Type B: The incorrect educational affiliation co-occurs with the senator name, and the incorrect fact is entailed in a sample of ten documents, (3) Type C: The name does not occur in any documents with the correct or hallucinated affiliation. We observe that the correct educational affiliations are commonly present in the c4 corpus for Llama models (**Type A error**, Fig. 4a).

5.2 Content-Grounded Tasks

Summarization In the task of abstractive summarization, statements in a generated summary that are not *faithful* to the provided context are considered as hallucinated, even if factually correct. In particular, we seek to understand if models hallucinations are caused by models incorrectly processing information in the input (*intrinsic hallucinations*), or by introducing information that cannot be inferred from the input (*extrinsic hallucinations*) (Maynez et al., 2020).

In order to study errors of the most capable models, we aggregate and examine the summaries of models whose utility score is at least 0.85. We manually annotate 100 statements in model summaries that were identified as hallucination, discarding cases where the entailment is ambiguous

Model	Examples	Corpus	Coverage
OLMo	libp2p_swarm, cryptomath, azdevclient, your_project_directory	Dolma	38.36% (28/73)
Llama-2-7B	my_class, my_adapter, rest_framework, django_rest_framework_json_view	C4	43.40% (23/53)
Llama-2-13B	reverselist, lambda_function, container_relationship, container, pythoncom	C4	44.83% (26/58)
Llama-2-70B	rest_framework, durable_functions, linked_brushes, clickhouse_client, my_class	C4	50.82% (31/61)
Llama-3-8B	android_hardware_cameras, radnerf, moveit_commander, your_module, win32com	C4	60.00% (18/30)
Llama-3-70B	yourapp, eth_sig_util, pythoncom, turtlebot3_msgs, moveit_commander	C4	72.41% (21/29)
GPT-3.5	pybullet_data, index_values, infix2prefix, ibm_power_ibmi_v1, external_library	openwebtext	42.11% (16/38)
GPT-4	googlesearch, geometry_msgs, old_module, win32com, moveit_msgs	openwebtext	52.00% (13/25)

Table 4: **Coverage of unique hallucinated packages found in pretraining data.** A considerable proportion of the hallucinated packages appear in the training data.

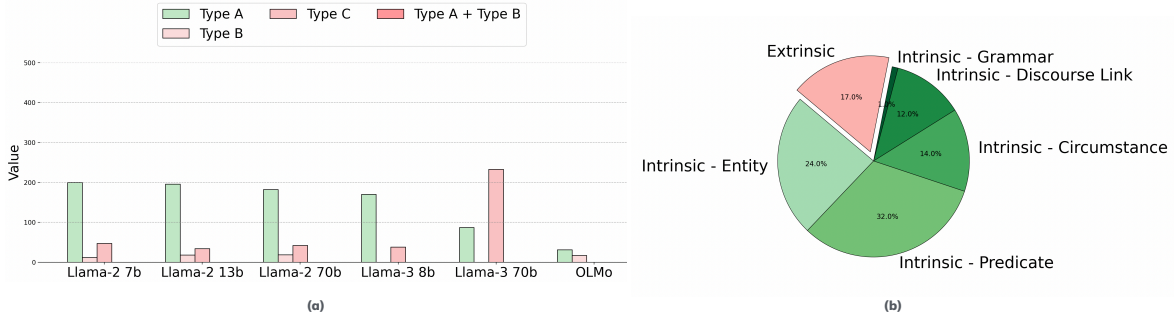


Figure 4: (a): **Counts of types of model hallucinations on educational affiliations of senators.** Models often hallucinate despite evidence of the correct fact within pretraining corpora. (b): **Distribution of hallucination types in model generations for a content-grounded task: abstractive summarization.** The vast majority of model hallucinations do not stem from the introduction of an external fact.

or where there was an error in atomization. We find that for high-utility models, **83% of model hallucinations are due to the model incorrectly processing the provided context (intrinsic hallucinations)**, with only 17% of errors originating from a model introducing an external fact into the summary. We further code each intrinsic hallucination with a fine-grained error category based on the typology introduced in (Pagnoni et al., 2021b). These categorize factuality errors as entity errors, relation error, errors of circumstance, coreference errors, discourse link errors, or grammatical errors (Fig. 4b). We find modern large language models seldom make grammatical errors, with incorrect entities or predicates being common sources of hallucination errors. Further, we find that most of the extrinsic hallucination errors originate from smaller models, with OLMo 7b instruct introducing 64.7% (11/17) of the extrinsic hallucination errors. On further coding 50 samples from OLMo 7b instruct, we find that extrinsic hallucinations account for 46% of its hallucination errors. However, we find that only 87% of these hallucinations contain an attributable fact, that these hallucinations often in-

troduce additional temporal information (30.4%), and that on sampling ten relevant documents from the pretraining data for each attributable fact, we are unable to find evidence of these hallucinations.

Simplification In order to study errors of most capable models, we aggregate and examine the simplified generations of models whose utility score is at least 0.85. We manually annotate 100 atomic statements in the automatically simplified texts that were identified as hallucination, discarding cases where the entailment is ambiguous or where there was an error in atomization. We categorize the hallucinations by type (inserting new factual information, substituting existing factual information, or deleting factual information in a way that introduces an unsupported fact), as well as severity, following the taxonomy proposed in (Devaraj et al., 2022a) for text simplification. Note that an atomic fact may feature multiple types of errors. First, we observe that 49% of samples feature insertion errors, 49% feature substitution errors, and 7% feature deletion errors. Moreover, 93.8% of the insertion errors are severe (introduce a new idea into